Supplementary Material

Which reverts count? Reverter populations, attribution rules and observation windows behind a Wikipedia indicator of information quality work

Tables are generated by 03_pipeline/17_supplement_v06.py from the aggregates in 04_output/ of the public repository. Editions marked * were held out for the registered tests. Dump rule: the history dumps' unbounded identity-revert rule. Radius rule: the same hashes with the 15-revision radius of the mwreverts library. Except for Table S7 and the registered rows of Table S8, all analyses are exploratory. Cells with fewer than 50 edits are not shown.

S1. Registration, sequence of events and deviations

The registered document (Zenodo, https://doi.org/10.5281/zenodo.23720117; commit 7fcff3d, 28 October 2026, 04:39 UTC) fixed the cohort, the three attribution rules, the reverter filters, the windows, four hypotheses with thresholds, and the code. The sequence was as follows.

1. English, Korean, French and German were analyzed without restriction. The thresholds of H1 to H4 were chosen from these four editions.
2. A batch driver downloaded and processed Japanese, Polish and Italian on 27 and 28 October 2026. The driver wrote its output files to disk and printed progress lines; the summary lines for H1 and H2 were among the lines printed to the terminal log. I did not open the output files or read those lines before the registered commit. This is recorded in the work log and by file and commit timestamps, and it cannot be verified by a reader.
3. The registration was committed and deposited. The results for the held-out editions were committed 11 minutes later, which reflects that they had already been computed.
4. Section 1 of the registered text says that the held-out editions "had not been downloaded". That sentence is wrong (deviation 4 below); the header of the same document states the actual sequence.

The registration is therefore a hold-out replication with thresholds fixed in advance on other editions. It protects against choosing thresholds after seeing the held-out results only to the extent that the reader accepts step 2.

Decisions. H1, H2 and H4 are tested with one-sided 95 percent bootstrap bounds (editors resampled), Holm-adjusted over the three tests within an edition. With three tests the Holm procedure compares the smallest p value with .05/3, the next with .05/2 and the largest with .05. All supported decisions in Table S7 have bootstrap p values below .0005 (no replicate on the wrong side of the threshold in 2,000), so the adjustment changes none; the one unsupported decision (H4, Polish) is unsupported without adjustment. H3 is a descriptive criterion without a test.

Deviations (full log: 04_logs/DEVIATION_LOG.md in the repository).

#	Registered	Done	Effect
1	Mixed model as an exploratory complement	Not reported; a per-edition logistic model was run and withdrawn as uninformative	None
2	Robustness list (12 h, 48 h, 30 d windows; five anchors; all registered accounts; historical-only bot flag; undo as a tool; 5-second latency; leave-one-edition-out; by year; Korean former anchor definition)	All run; reported in Table S8. Added: 180-day window, current-only bot flag, exclusion of unregistered reverters, cohort from 2021, Korean surge split	None
3	H3, second clause	Rejected in all three editions and reported as rejected; mechanism examined afterwards (exploratory)	Decision unchanged
4	Text: held-out editions “had not been downloaded”	They had been downloaded and processed; outputs unopened (see above)	Weakens the evidential status of the tests
5	2,000 bootstrap replicates	Registered code used 2,000 for H1 and H2, 500 for H4; H4 recomputed with 2,000	No decision changes
6	Negative control; simulation	Withdrawn: the control cannot fail and the simulation restates a closed form. Outputs remain in the repository	None
7	Interpretation of H2 as human substitution for automation	Withdrawn. Follow-up reverts redefined as reverts by a third party; radius rule, reverter classes, retention by class added (exploratory)	Decisions unchanged; interpretation changed
8	Tag analyses for first edits from August 2020	Coverage analyses restricted to first edits from October 2020 (tag enabled on all wikis 15 September 2020); H3 reported with the registered window	First clause unaffected. Second clause: rejected in all three editions with the registered window; with the October window it would be met in Japanese (Table S10)
9	“Successor filters” F0, F1, F2	Reverter classes; tool exclusion presented as hypothetical	None
10	Release of editor-level derived data with randomized identifiers	Not released: one first edit per editor with its timestamp and edition re-identifies the account from the public history whatever the identifier. Code regenerates the files	Reduces direct reusability; aggregates and code are public

#	Registered	Done	Effect
11	Minimum cell size 50	Applied in all tables of manuscript v0.6; an earlier public draft showed some smaller cells	None
12	Holm adjustment “across H1 to H4”	Applied over H1, H2 and H4; H3 has no test statistic. H1: all three R values are at or above the registered .75, read as registered (“most of the filter effect persists”). H4: registered as a point ratio of at least 1.5 with a lower bound above 1.0; reported so	None
13	Model used for coding assistance	The review coding by a language model and its checks are described in the manuscript; an internal pre-submission check of drafts by language-model reviewers led to several of the changes above and is recorded in the work log	None

Table S1. Structured review of practice: how 41 studies using a revert-based measure operationalize it

One row per study. “Not stated” means that no sentence establishing the choice was found in the article or its supplement. The full table with the establishing sentences, URLs, evidence levels, the corrections made on re-verification and the second-pass codes is 04_output/practice_review.csv and the files beside it. Search record: 04_output/practice_review_LIS_search.md.

Flow of studies. Stage 1 (citation networks of the methodological works and keyword searches; the number of records screened was not logged): 36 eligible studies, of which 34 were coded and tallied and 2 are listed but not tallied because the full text could not be read. Stage 2 (OpenAlex, information science and neighboring journals): 911 records from six queries and 258 from a title search (1,169 in all); 79 assessed for eligibility; 40 not assessable because the full text was not accessible; 8 eligible, of which 7 were coded and tallied and 1 (an earlier conference report of the same sample as Stvilia et al., 2007) is listed but not tallied. Total tallied: 41.

Coding procedure. First-pass and re-check coding were done with Claude (Anthropic; model versions and dates in 04_logs/AI_ASSISTANCE_LOG.md), instructed to code a choice as stated only on finding an establishing passage and to record that passage. The notes and evidence_level columns record, per study, where a code rests on an extracted passage or an earlier extract and not on a re-read of the full text. No human second coder was used.

short_cite	venue	editions	construct	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	source
Kittur et al., 2007	CHI 2007	en only	conflict	identity + summary matching	none stated	not stated	not stated	excluded	not stated	partial	citation-network search
Priedhorsky et al., 2007	GROUP '07	en only	quality judgment	identity + summary matching	15 revisions	counted explicitly	counted explicitly	not stated	revision-bounded: 15 revisions	partial	citation-network search
Halfaker et al., 2009	WikiSym '09	en only	rejection	identity hash	none stated	not stated	not stated	excluded	unbounded stated	yes: vandalism reverts removed	citation-network search
Suh et al., 2009	WikiSym '09	en only	rejection	summary matching only	none stated	not stated	not stated	not stated	not stated	no	citation-network search
Ekstrand & Riedl, 2009	WikiSym '09	en only	other	identity hash	none stated	not stated	not stated	not stated	not stated	no	citation-network search
West et al., 2010	EUROSEC '10	en only	quality judgment	rollback or undo actions only	none stated	excluded	counted explicitly	not stated	not stated	no	citation-network search
Halfaker et al., 2011a	WikiSym '11	en only	rejection	not stated	none stated	not stated	not stated	excluded	not stated	no	citation-network search
Halfaker et al., 2011b	WikiSym '11	en only	rejection	rollback or undo actions only	none stated	not stated	not stated	not stated	not stated	no	citation-network search
Lam et al., 2011	WikiSym '11	en only	rejection	identity + summary matching	none stated	not stated	not stated	not stated	time-bounded: 7 days	no	citation-network search
Sumi et al., 2011	IEEE PASSAT/SocialCom 2011	multilingual	conflict	identity hash	none stated	not stated	not stated	excluded	not stated	partial	citation-network search
Yasseri et al., 2012	PLoS ONE	en only	conflict	identity hash	none stated	not stated	not stated	excluded	not stated	no	citation-network search
Flöck et al., 2012	HT '12	en only	other	diff-based	20 revisions (also 50, 100)	not stated	not stated	not stated	revision-bounded: 20 revisions	yes: scan range 20/50/100; diff vs identity	citation-network search
Halfaker et al., 2013	American Behavioral Scientist	en only	rejection	identity hash	none stated	modelled separately or focal	modelled separately	excluded (definitional)	not stated	no	citation-network search
Geiger & Halfaker, 2013	WikiSym '13	en only	effort	not stated	none stated	modelled separately or focal	modelled separately	not stated	unbounded stated	no	citation-network search

short_cite	venue	editions	construct	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	source
Yasseri et al., 2014	Global Wikipedia (edited book)	multilingual	conflict	identity hash	none stated	not stated	not stated	not stated	not stated	no	citation-network search
Tsvetkova et al., 2016	Scientific Reports	multilingual	conflict	identity hash	none stated	excluded	not stated	excluded	not stated	no	citation-network search
Tsvetkova et al., 2017	PLoS ONE	multilingual	conflict	identity hash	none stated	modelled separately or focal	not stated	excluded	not stated	no	citation-network search
Geiger & Halfaker, 2017	PACM HCI (CSCW)	multilingual	conflict	identity hash	mwreverts default (not reported)	modelled separately or focal	not stated	not stated	not stated	partial	citation-network search
TeBlunthuis et al., 2018	CHI 2018	Wikia	rejection	not stated	none stated	modelled separately or focal	not stated	not stated	not stated	no	citation-network search
DeDeo, 2013	PLoS ONE	en only	conflict	identity hash	none stated	not stated	not stated	excluded	not stated	partial	citation-network search
Zhang et al., 2019	ICWSM 2019	en only	rejection	not stated	none stated	not stated	not stated	excluded (definitional)	unbounded stated	no	citation-network search
Tran et al., 2020	IEEE S&P 2020	en only	quality judgment	identity hash	mwreverts default (not reported)	not stated	not stated	excluded	not stated	partial	citation-network search
Hill & Shaw, 2021	Communication Research	Wikia	quality judgment	identity hash	mwreverts default (not reported)	not stated	not stated	not stated	not stated	no	citation-network search
Chhabra et al., 2020	OpenSym 2020	en only	conflict	identity hash	none stated	not stated	not stated	counted explicitly	not stated	no	citation-network search
TeBlunthuis et al., 2021	PACM HCI (CSCW)	multilingual	rejection	identity hash	none stated	not stated	not stated	excluded	time-bounded: 48 hours	no	citation-network search
Ruprechter et al., 2021	Scientific Reports	multilingual	other	dump precomputed field	none stated	counted explicitly	not stated	not stated	not stated	no	citation-network search
Wikimedia Foundation, 2021	Meta-Wiki report	single non-English edition (pt)	quality judgment	not stated	none stated	not stated	not stated	not stated	time-bounded: 48 hours	no	citation-network search
Tran et al.,	PACM HCI	multilingual	quality	not stated	none stated	not stated	not stated	not stated	unbounded stated	no	citation-network

short_cite	venue	editions	construct	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	source
2022	(CSCW)		judgment								search
Wang & Zhu, 2022	FAccT '22	multilingual	effort	not stated	none stated	not stated	not stated	excluded (definitional)	not stated	no	citation-network search
Champion & Hill, 2023	PACMHCI (CSCW)	en only	quality judgment	not stated	10 contributions forward	not stated	not stated	not stated	revision-bounded: 10 contributions	no	citation-network search
Warncke-Wang et al., 2023	PACMHCI (CSCW)	multilingual	quality judgment	not stated	none stated	not stated	not stated	not stated	time-bounded: 48 hours	no	citation-network search
Trokhymovych et al., 2023	KDD '23	multilingual	quality judgment	dump precomputed field	none stated	not stated	not stated	not stated	not stated	no	citation-network search
Franzmeyer et al., 2024	Findings of ACL 2024	en only	quality judgment	not stated	none stated	not stated	not stated	excluded (definitional)	not stated	no	citation-network search
Formisano et al., 2026	Humanities and Social Sciences Communications	en only	effort	dump precomputed field	none stated	not stated	not stated	not stated	not stated	no	citation-network search
Stvilia et al., 2007	Journal of the American Society for Information Science and Technology	en only	quality judgment	not stated	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal search
Purdy, 2020	First Monday	en only	other	diff-based	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal search
Iñiguez et al., 2014	EPJ Data Science	multilingual	conflict	identity hash	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal search
Rudas et al., 2017	PLoS ONE	multilingual	conflict	not stated	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal search
Lerner & Lomi, 2018	PLoS ONE	en only	other	not stated	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal search

short_cite	venue	editions	construct	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	source
Yasseri & Kertész, 2013	Journal of Statistical Physics	multilingual	conflict	identity hash	none stated	excluded	not stated	not stated	not stated	partial	LIS and neighbouring journal search
Sepehri Rad & Barbosa, 2012	WikiSym ’12	not stated	conflict	not stated	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal search

Checklist items 1 to 4 (Table 5 of the article) applied to the reviewed studies. Item 1 (construct) could be assigned by the coder for all studies but is seldom named by the study itself; item 2 (attribution rule) is stated in 27 of 41; item 3 (window) in 11; item 4 (bots, tools and self-reverts, all three) in 1, at least one of the three in 22.

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Table S2. Composition of first reverts of newcomer first edits (dump rule, full cohorts)

First edits 2019 onward, window 24 h. Shares of first reverts; intervals are 95 percent percentile intervals from 2,000 bootstrap resamples of reverting accounts (each revert by an unregistered editor counted as its own account).

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	2,189,284	0.37	0.122 [0.001, 0.300]	0.448 [0.346, 0.534]	0.364 [0.286, 0.432]	0.023 [0.018, 0.027]	0.044 [0.034, 0.053]
Italian*	135,400	0.342	0.000 [0.000, 0.001]	0.525 [0.455, 0.583]	0.407 [0.354, 0.468]	0.024 [0.019, 0.032]	0.044 [0.035, 0.058]
Korean	47,633	0.216	0.003 [0.000, 0.008]	0.516 [0.349, 0.618]	0.289 [0.232, 0.370]	0.096 [0.068, 0.147]	0.096 [0.068, 0.146]
German	236,903	0.256	0.001 [0.000, 0.003]	0.410 [0.351, 0.466]	0.545 [0.492, 0.599]	0.017 [0.014, 0.020]	0.027 [0.023, 0.033]
Polish*	67,755	0.222	0.001 [0.000, 0.004]	0.315 [0.263, 0.374]	0.616 [0.556, 0.668]	0.030 [0.023, 0.041]	0.038 [0.029, 0.050]
French	346,938	0.243	0.042 [0.000, 0.125]	0.255 [0.189, 0.324]	0.643 [0.564, 0.719]	0.012 [0.010, 0.015]	0.047 [0.039, 0.058]
Japanese*	223,045	0.14	0.001 [0.000, 0.002]	0.247 [0.146, 0.339]	0.558 [0.491, 0.631]	0.080 [0.066, 0.098]	0.114 [0.094, 0.140]

First edits 2019 onward, window unbounded. Shares of first reverts; intervals are 95 percent percentile intervals from 2,000 bootstrap resamples of reverting accounts (each revert by an unregistered editor counted as its own account).

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	2,189,284	0.412	0.111 [0.001, 0.274]	0.432 [0.341, 0.508]	0.383 [0.308, 0.448]	0.034 [0.027, 0.040]	0.040 [0.032, 0.048]
Italian*	135,400	0.369	0.001 [0.000, 0.002]	0.503 [0.428, 0.561]	0.421 [0.369, 0.482]	0.034 [0.027, 0.044]	0.041 [0.033, 0.053]

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
Korean	47,633	0.279	0.006 [0.000, 0.015]	0.438 [0.286, 0.544]	0.336 [0.282, 0.405]	0.142 [0.103, 0.206]	0.078 [0.055, 0.114]
German	236,903	0.294	0.003 [0.000, 0.007]	0.369 [0.314, 0.427]	0.586 [0.533, 0.637]	0.017 [0.014, 0.021]	0.025 [0.021, 0.030]
Polish*	67,755	0.303	0.002 [0.000, 0.005]	0.248 [0.197, 0.306]	0.690 [0.627, 0.745]	0.031 [0.024, 0.041]	0.029 [0.022, 0.038]
French	346,938	0.27	0.039 [0.001, 0.116]	0.241 [0.177, 0.305]	0.654 [0.583, 0.725]	0.023 [0.019, 0.027]	0.044 [0.036, 0.053]
Japanese*	223,045	0.172	0.003 [0.000, 0.007]	0.209 [0.124, 0.295]	0.581 [0.518, 0.643]	0.109 [0.091, 0.130]	0.097 [0.081, 0.115]

First edits 2021 onward, window 24 h. Shares of first reverts; intervals are 95 percent percentile intervals from 2,000 bootstrap resamples of reverting accounts (each revert by an unregistered editor counted as its own account).

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	1,453,656	0.361	0.125 [0.001, 0.304]	0.524 [0.407, 0.620]	0.285 [0.223, 0.346]	0.022 [0.017, 0.027]	0.044 [0.034, 0.054]
Italian*	88,005	0.334	0.000 [0.000, 0.000]	0.546 [0.467, 0.611]	0.390 [0.331, 0.457]	0.020 [0.015, 0.028]	0.044 [0.033, 0.060]
Korean	32,751	0.239	0.001 [0.000, 0.003]	0.554 [0.389, 0.648]	0.262 [0.206, 0.345]	0.094 [0.065, 0.148]	0.090 [0.062, 0.140]
German	159,216	0.248	0.001 [0.000, 0.003]	0.394 [0.332, 0.456]	0.560 [0.500, 0.617]	0.018 [0.015, 0.022]	0.028 [0.023, 0.034]
Polish*	45,382	0.227	0.002 [0.000, 0.005]	0.303 [0.244, 0.368]	0.624 [0.562, 0.679]	0.035 [0.025, 0.048]	0.037 [0.027, 0.049]
French	229,556	0.232	0.020 [0.000, 0.062]	0.261 [0.194, 0.325]	0.656 [0.595, 0.721]	0.011 [0.009, 0.013]	0.051 [0.042, 0.062]
Japanese*	148,508	0.138	0.001 [0.000, 0.003]	0.300 [0.171, 0.411]	0.519 [0.441, 0.612]	0.069 [0.054, 0.089]	0.110 [0.086, 0.141]

First edits 2021 onward, window unbounded. Shares of first reverts; intervals are 95 percent percentile intervals from 2,000 bootstrap resamples of reverting accounts (each revert by an unregistered editor counted as its own account).

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	1,453,656	0.402	0.114 [0.001, 0.280]	0.508 [0.401, 0.594]	0.305 [0.243, 0.364]	0.033 [0.026, 0.039]	0.040 [0.032, 0.049]
Italian*	88,005	0.361	0.001 [0.000, 0.001]	0.522 [0.445, 0.583]	0.406 [0.351, 0.471]	0.030 [0.023, 0.040]	0.041 [0.032, 0.056]
Korean	32,751	0.303	0.004 [0.000, 0.009]	0.477 [0.307, 0.581]	0.310 [0.256, 0.386]	0.135 [0.094, 0.207]	0.074 [0.052, 0.115]
German	159,216	0.287	0.003 [0.000, 0.007]	0.353 [0.297, 0.415]	0.600 [0.542, 0.653]	0.018 [0.015, 0.022]	0.025 [0.021, 0.030]
Polish*	45,382	0.316	0.002 [0.000, 0.005]	0.234 [0.174, 0.299]	0.704 [0.636, 0.767]	0.033 [0.024, 0.045]	0.027 [0.020, 0.037]
French	229,556	0.26	0.019 [0.001, 0.059]	0.246 [0.184, 0.307]	0.667 [0.607, 0.728]	0.020 [0.017, 0.025]	0.047 [0.040, 0.057]

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
Japanese*	148,508	0.17	0.004 [0.000, 0.010]	0.254 [0.145, 0.361]	0.553 [0.477, 0.634]	0.096 [0.077, 0.119]	0.093 [0.074, 0.115]

Concentration of reverting, first edits 2019 onward, 24 hours.

Edition	Bot: accounts	Bot: share of the most active account	One-click tool: accounts	One-click tool: share of the most active account	Other registered, manual: accounts	Other registered, manual: share of the most active account
English	29	0.931	6,596	0.061	22,356	0.017
Italian*	6	0.417	201	0.064	1,100	0.068
Korean	5	0.484	93	0.187	521	0.09
German	6	0.395	1,286	0.09	2,615	0.053
Polish*	2	0.789	341	0.109	726	0.101
French	10	0.981	316	0.125	2,756	0.039
Japanese*	5	0.714	103	0.234	2,285	0.045

First edits that create a page (page-history samples). Such edits cannot be identity-reverted; removal is by deletion, which is outside the indicator. English restricts article creation to autoconfirmed accounts.

Edition	Share of first edits that create a page	p reverted within 24 h, all	p reverted within 24 h, edits to existing pages	First reverts that undo one revision only
English	0	0.367	0.367	0.731
Italian*	0.155	0.341	0.403	0.72
Korean	0.145	0.216	0.252	0.643
German	0.16	0.256	0.304	0.788
Polish*	0.201	0.221	0.277	0.719
French	0.121	0.243	0.277	0.721
Japanese*	0.093	0.14	0.155	0.672

Table S3. Change in the revert rate when a class of first reverts is recoded as not reverted, and what the tool class can see

Percentage points; by identity, the class's share of first reverts times the rate.

First edits 2019 onward, 24-hour window.

Edition	Bots	One-click tools	Self-reverts	Unregistered reverters	Undo-button reverts (if counted as a tool)	Undo share of manual reverts by other registered editors
English	4.53	16.56	1.62	0.84	11.24	0.784
Italian*	0.01	17.93	1.49	0.83	10.17	0.699
Korean	0.07	11.13	2.07	2.07	5.87	0.71
German	0.04	10.5	0.69	0.43	7.31	0.489
Polish*	0.03	6.99	0.83	0.67	7.54	0.502
French	1.02	6.2	1.15	0.3	11.69	0.722
Japanese*	0.01	3.45	1.6	1.11	5.69	0.618

First edits 2021 onward, 24-hour window.

Edition	Bots	One-click tools	Self-reverts	Unregistered reverters	Undo-button reverts (if counted as a tool)	Undo share of manual reverts by other registered editors
English	4.51	18.9	1.58	0.79	8.49	0.755
Italian*	0	18.21	1.46	0.67	9.76	0.725
Korean	0.03	13.25	2.15	2.24	5.97	0.71
German	0.03	9.76	0.68	0.45	7.19	0.48
Polish*	0.04	6.87	0.83	0.79	8.37	0.532
French	0.48	6.07	1.19	0.26	11.81	0.749
Japanese*	0.02	4.15	1.52	0.96	5.46	0.645

Wider tool tags. Share of first reverts within 24 hours carrying a tag of RedWarn, Ultraviolet, AntiVandal script, STiki, LiveRC or MassRollback but outside the tool class used in the article (rollback, Huggle, Twinkle). Adding them would raise the tool share by the amount shown.

Edition	First edits	First reverts within 24 h	Tool class as used	Wider tags, outside the tool class
English	2019+	809720	0.448	0.039
English	2021+	524398	0.524	0.049
Italian*	2019+	46260	0.525	0.000
Italian*	2021+	29357	0.546	0.000
Korean	2019+	10279	0.516	0.000
Korean	2021+	7838	0.554	0.000
German	2019+	60665	0.410	0.000
German	2021+	39495	0.394	0.000
Polish*	2019+	15033	0.315	0.000
Polish*	2021+	10281	0.303	0.000
French	2019+	84351	0.255	0.085
French	2021+	53369	0.261	0.080
Japanese*	2019+	31127	0.247	0.000
Japanese*	2021+	20515	0.300	0.000

Tool-related change tags on reverting revisions in the article namespace, whole period, by edition (tags matching rollback, Huggle, Twinkle, RedWarn, Ultraviolet, STiki, AntiVandal, SWViewer, LiveRC and similar names; tags not listed for an edition do not occur there).

Edition	Tag	Reverting revisions
Korean	mw-rollback	218443
Korean	app-rollback	23
Japanese*	mw-rollback	160250
Japanese*	MassRollback	19619
Japanese*	app-rollback	11
Polish*	mw-rollback	290595
Polish*	app-rollback	109
Italian*	mw-rollback	1082706
Italian*	app-rollback	240
German	mw-rollback	1139847
German	app-rollback	400

Edition	Tag	Reverting revisions
French	mw-rollback	587008
French	LiveRC	285273
French	huggle	28943
French	app-rollback	68
English	mw-rollback	6999688
English	twinkle	3271154
English	huggle	1210355
English	RedWarn	433575
English	Ultraviolet	157322
English	AntiVandal script	156948
English	STiki	97797
English	app-rollback	1240

Table S4. Follow-up manual reverts by a third party, and the window ratio R, under two attribution rules (page-history samples)

Share of first edits first reverted by the class shown that were later reverted manually (no bot flag, no tool tag) by a registered account that is neither the author nor the account that made the first revert. The window is counted from the first revert. Unregistered follow-up reverters are not counted, because they cannot be compared with the first reverter; tables that count them are marked.

Dump rule, 24 h.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.077	0.079	0.069	0.057	0.060	0.090	0.097
Manual, not the author	0.066	0.084	0.064	0.041	0.037	0.070	0.092
Bot	0.126		0.078	0.063		0.217	
One-click tool	0.064	0.079	0.069	0.057	0.060	0.070	0.098
Other registered, manual	0.066	0.082	0.068	0.041	0.035	0.070	0.091
Unregistered	0.057	0.117	0.056	0.070	0.077	0.061	0.094
Self	0.079	0.084	0.155	0.099	0.100	0.073	0.076

Dump rule, 7 d.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.117	0.133	0.135	0.084	0.098	0.138	0.128
Manual, not the author	0.110	0.133	0.115	0.064	0.065	0.111	0.126
Bot	0.173		0.169	0.063		0.271	
One-click tool	0.103	0.133	0.135	0.084	0.099	0.116	0.129
Other registered, manual	0.111	0.131	0.119	0.063	0.062	0.111	0.125
Unregistered	0.096	0.156	0.103	0.085	0.119	0.097	0.136
Self	0.117	0.143	0.206	0.118	0.125	0.098	0.092

Dump rule, unbounded.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.182	0.229	0.272	0.160	0.220	0.241	0.203
Manual, not the author	0.174	0.211	0.212	0.125	0.154	0.196	0.187
Bot	0.243		0.273	0.159		0.363	
One-click tool	0.166	0.230	0.272	0.160	0.221	0.222	0.204
Other registered, manual	0.176	0.209	0.218	0.124	0.152	0.197	0.185
Unregistered	0.151	0.236	0.200	0.170	0.200	0.174	0.195
Self	0.165	0.218	0.286	0.202	0.204	0.175	0.132

Dump rule, unbounded: follow-up by any third party, registered or not, after the first edit's content had been reinstated (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.014	0.021	0.122	0.017	0.028	0.010	0.042
Manual, not the author	0.016	0.021	0.070	0.016	0.014	0.013	0.042

Dump rule, unbounded: follow-up by any third party, registered or not, immediately, restoring a still earlier state (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.011	0.015	0.024	0.010	0.016	0.008	0.007
Manual, not the author	0.011	0.015	0.024	0.004	0.007	0.008	0.014

Dump rule, unbounded: follow-up by any third party, registered or not, after other intervening edits (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.176	0.217	0.208	0.145	0.191	0.234	0.177
Manual, not the author	0.175	0.204	0.240	0.114	0.146	0.188	0.178

Radius rule, 24 h.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.010	0.013	0.017	0.009	0.013	0.013	0.029
Manual, not the author	0.006	0.011	0.012	0.003	0.003	0.005	0.012
Bot	0.017		0.026	0.032		0.044	
One-click tool	0.008	0.013	0.016	0.009	0.013	0.008	0.029
Other registered, manual	0.006	0.010	0.010	0.003	0.002	0.005	0.013
Unregistered	0.011	0.030	0.015	0.004	0.025	0.006	0.009
Self	0.019	0.029	0.064	0.052	0.058	0.024	0.032

Radius rule, 7 d.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.013	0.016	0.023	0.011	0.017	0.016	0.031

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Manual, not the author	0.009	0.013	0.016	0.003	0.004	0.006	0.017
Bot	0.020		0.039	0.032		0.048	
One-click tool	0.011	0.016	0.023	0.010	0.017	0.010	0.032
Other registered, manual	0.008	0.011	0.013	0.003	0.003	0.006	0.017
Unregistered	0.014	0.031	0.022	0.004	0.032	0.006	0.015
Self	0.026	0.035	0.072	0.057	0.060	0.029	0.035

Radius rule, unbounded.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.017	0.019	0.032	0.012	0.023	0.019	0.035
Manual, not the author	0.011	0.016	0.022	0.004	0.006	0.008	0.021
Bot	0.026		0.053	0.048		0.052	
One-click tool	0.014	0.019	0.032	0.012	0.023	0.014	0.035
Other registered, manual	0.011	0.014	0.019	0.004	0.005	0.008	0.022
Unregistered	0.014	0.036	0.029	0.004	0.034	0.008	0.019
Self	0.031	0.039	0.077	0.064	0.066	0.031	0.038

Radius rule, unbounded: follow-up by any third party, registered or not, after the first edit's content had been reinstated (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.001	0.001	0.009	0.001	0.002	0.000	0.004
Manual, not the author	0.001	0.002	0.007	0.000	0.001	0.000	0.002

Radius rule, unbounded: follow-up by any third party, registered or not, immediately, restoring a still earlier state (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.010	0.014	0.020	0.010	0.016	0.008	0.007
Manual, not the author	0.011	0.014	0.018	0.004	0.007	0.008	0.013

Radius rule, unbounded: follow-up by any third party, registered or not, after other intervening edits (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.011	0.012	0.023	0.005	0.010	0.013	0.029
Manual, not the author	0.006	0.011	0.016	0.002	0.004	0.005	0.015

Dump rule, 24 hours, by the number of revisions on the page before the measured edit (fixed before the first revert).

key	English	Italian*	Korean	German	Polish*	French	Japanese*
1. under 50, Bot or one-click tool	0.056	0.045	0.046	0.045	0.049	0.057	0.069
1. under 50, Other registered, manual	0.039	0.053	0.047	0.036	0.023	0.039	0.061
2. 50 to 499, Bot or one-click tool	0.078	0.071	0.066	0.061	0.059	0.082	0.097
2. 50 to 499, Other registered, manual	0.075	0.074	0.068	0.037	0.035	0.065	0.091
3. 500 or more, Bot or one-click tool	0.103	0.101	0.113	0.061	0.074	0.115	0.126
3. 500 or more, Other registered, manual	0.092	0.101	0.092	0.056	0.061	0.092	0.120

Radius rule, 24 hours, by the number of revisions on the page before the measured edit (fixed before the first revert).

key	English	Italian*	Korean	German	Polish*	French	Japanese*
1. under 50, Bot or one-click tool	0.007	0.007	0.008	0.008	0.010	0.009	0.014
1. under 50, Other registered, manual	0.002	0.007	0.004	0.003	0.000	0.002	0.006
2. 50 to 499, Bot or one-click tool	0.010	0.011	0.013	0.009	0.012	0.013	0.035
2. 50 to 499, Other registered, manual	0.007	0.007	0.012	0.003	0.002	0.005	0.016
3. 500 or more, Bot or one-click tool	0.016	0.017	0.038	0.009	0.018	0.014	0.031
3. 500 or more, Other registered, manual	0.009	0.014	0.014	0.003	0.005	0.007	0.013

Dump rule, 24 hours counted from the measured edit (the registered clock; follow-up by any third party, registered or not).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.086	0.088	0.090	0.063	0.069	0.096	0.106
Manual, not the author	0.087	0.103	0.140	0.053	0.056	0.078	0.142
Bot	0.135					0.227	
One-click tool	0.073	0.088	0.089	0.063	0.069	0.074	0.107
Other registered, manual	0.080	0.095	0.113	0.050	0.052	0.076	0.126
Unregistered	0.200	0.237	0.222	0.131	0.156	0.175	0.259
Self	0.092	0.104	0.197	0.117	0.115	0.082	0.084

Window ratio R (replace effect of excluding bots and tools at the longer window divided by the effect within 24 hours; 2,000 bootstrap resamples of editors). With N(W) the number of first edits first reverted by a bot or tool within W and S(W) the share of them with a later non-bot, non-tool revert within W, the replace effect is N(W)(1 − S(W))/n, so R = N(∞)(1 − S(∞)) / [N(24

h)(1 − S(24 h))]. R falls below 1 when follow-up accumulates faster than first reverts by bots and tools, and rises above 1 when it does not.

key	English	Italian*	Korean	German	Polish*	French	Japanese*
Dump rule, 180 d / 24 h	0.926 [0.919, 0.934]	0.857 [0.848, 0.865]	0.809 [0.793, 0.825]	0.931 [0.921, 0.941]	0.905 [0.890, 0.921]	0.875 [0.861, 0.889]	0.926 [0.906, 0.945]
Dump rule, unbounded / 24 h	0.922 [0.915, 0.930]	0.843 [0.834, 0.852]	0.775 [0.758, 0.792]	0.917 [0.907, 0.928]	0.877 [0.860, 0.892]	0.858 [0.844, 0.871]	0.919 [0.898, 0.939]
Radius rule, 180 d / 24 h	1.043 [1.038, 1.047]	1.024 [1.019, 1.028]	1.074 [1.064, 1.085]	1.036 [1.030, 1.042]	1.067 [1.057, 1.077]	1.039 [1.032, 1.046]	1.046 [1.034, 1.058]
Radius rule, unbounded / 24 h	1.045 [1.041, 1.050]	1.024 [1.020, 1.028]	1.074 [1.064, 1.085]	1.036 [1.031, 1.042]	1.067 [1.057, 1.077]	1.039 [1.032, 1.046]	1.048 [1.037, 1.060]

Table S5. Retention (an edit on days 2 to 60) by who made the first revert within 24 hours

Cells: share retained; odds ratio against newcomers not reverted within 24 hours with 95 percent interval; number of newcomers.

All newcomers.

first_revert_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Any revert	0.097; OR 0.43 [0.43, 0.43]; n = 809,720	0.090; OR 0.40 [0.38, 0.41]; n = 46,260	0.079; OR 0.32 [0.29, 0.34]; n = 10,279	0.107; OR 0.51 [0.50, 0.53]; n = 60,665	0.113; OR 0.54 [0.51, 0.57]; n = 15,033	0.089; OR 0.47 [0.46, 0.49]; n = 84,351	0.116; OR 0.36 [0.35, 0.37]; n = 31,127
Bot	0.069; OR 0.30 [0.29, 0.31]; n = 99,135			0.081; OR 0.40 [0.19, 0.85]; n = 86		0.041; OR 0.21 [0.18, 0.25]; n = 3,552	
One-click tool	0.086; OR 0.38 [0.37, 0.38]; n = 362,541	0.072; OR 0.31 [0.29, 0.33]; n = 24,278	0.035; OR 0.13 [0.12, 0.16]; n = 5,301	0.062; OR 0.29 [0.27, 0.30]; n = 24,876	0.067; OR 0.30 [0.27, 0.34]; n = 4,733	0.052; OR 0.27 [0.25, 0.28]; n = 21,513	0.037; OR 0.10 [0.09, 0.12]; n = 7,690
Other registered, manual	0.114; OR 0.52 [0.51, 0.52]; n = 294,376	0.108; OR 0.48 [0.46, 0.51]; n = 18,823	0.118; OR 0.50 [0.44, 0.56]; n = 2,971	0.142; OR 0.71 [0.68, 0.73]; n = 33,055	0.135; OR 0.66 [0.62, 0.70]; n = 9,263	0.104; OR 0.57 [0.55, 0.58]; n = 54,268	0.129; OR 0.41 [0.39, 0.43]; n = 17,362
Unregistered	0.106; OR 0.48 [0.45, 0.50]; n = 18,309	0.121; OR 0.55 [0.46, 0.66]; n = 1,130	0.122; OR 0.52 [0.43, 0.63]; n = 988	0.084; OR 0.40 [0.32, 0.50]; n = 1,007	0.132; OR 0.65 [0.50, 0.85]; n = 454	0.082; OR 0.44 [0.35, 0.55]; n = 1,044	0.152; OR 0.49 [0.44, 0.55]; n = 2,485
Self	0.139; OR 0.65 [0.63, 0.67]; n = 35,359	0.128; OR 0.59 [0.51, 0.67]; n = 2,017	0.153; OR 0.67 [0.56, 0.79]; n = 988	0.113; OR 0.55 [0.47, 0.64]; n = 1,641	0.124; OR 0.61 [0.47, 0.78]; n = 564	0.119; OR 0.66 [0.60, 0.73]; n = 3,974	0.199; OR 0.68 [0.63, 0.74]; n = 3,562

Share retained among newcomers not reverted within 24 hours: English 0.199; Italian* 0.201; Korean 0.213; German 0.189; Polish* 0.191; French 0.171; Japanese* 0.266.

Newcomers not blocked within 60 days of the first edit.

first_revert_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Any revert	0.096; OR 0.43 [0.43, 0.43]; n = 785,031	0.088; OR 0.40 [0.38, 0.41]; n = 41,280	0.080; OR 0.32 [0.30, 0.35]; n = 9,508	0.110; OR 0.53 [0.51, 0.54]; n = 57,461	0.112; OR 0.54 [0.51, 0.57]; n = 14,302	0.090; OR 0.48 [0.47, 0.50]; n = 78,726	0.112; OR 0.35 [0.34, 0.36]; n = 28,635
Bot	0.069; OR 0.30 [0.29, 0.31]; n = 95,666			0.081; OR 0.40 [0.19, 0.85]; n = 86		0.050; OR 0.26 [0.22, 0.31]; n = 2,662	
One-click tool	0.085; OR 0.38 [0.37, 0.38]; n = 352,569	0.069; OR 0.30 [0.29, 0.32]; n = 21,337	0.035; OR 0.14 [0.12, 0.16]; n = 4,806	0.066; OR 0.30 [0.29, 0.32]; n = 22,418	0.067; OR 0.31 [0.27, 0.34]; n = 4,280	0.053; OR 0.27 [0.26, 0.29]; n = 19,442	0.031; OR 0.09 [0.08, 0.10]; n = 7,052
Other registered, manual	0.113; OR 0.51 [0.51, 0.52]; n = 284,043	0.107; OR 0.49 [0.46, 0.51]; n = 17,054	0.119; OR 0.50 [0.45, 0.56]; n = 2,797	0.140; OR 0.70 [0.68, 0.72]; n = 32,379	0.132; OR 0.65 [0.61, 0.69]; n = 9,028	0.104; OR 0.57 [0.55, 0.59]; n = 51,753	0.125; OR 0.40 [0.38, 0.42]; n = 15,857
Unregistered	0.104; OR 0.47 [0.45, 0.49]; n = 17,863	0.126; OR 0.59 [0.49, 0.71]; n = 979	0.124; OR 0.53 [0.44, 0.65]; n = 916	0.087; OR 0.41 [0.33, 0.52]; n = 961	0.137; OR 0.68 [0.52, 0.90]; n = 438	0.081; OR 0.44 [0.35, 0.55]; n = 983	0.144; OR 0.47 [0.42, 0.53]; n = 2,275
Self	0.137; OR 0.64 [0.62, 0.66]; n = 34,890	0.126; OR 0.59 [0.52, 0.68]; n = 1,898	0.148; OR 0.65 [0.54, 0.77]; n = 960	0.114; OR 0.55 [0.48, 0.65]; n = 1,617	0.128; OR 0.63 [0.49, 0.81]; n = 538	0.119; OR 0.66 [0.60, 0.73]; n = 3,886	0.199; OR 0.69 [0.63, 0.75]; n = 3,423

Share retained among newcomers not reverted within 24 hours: English 0.198; Italian* 0.196; Korean 0.212; German 0.189; Polish* 0.190; French 0.169; Japanese* 0.265.

Share of newcomers blocked within 60 days of the first edit, by class of first revert within 24 hours.

first_revert_class	English	Italian*	Korean	German	Polish*	French	Japanese*
All newcomers	0.015	0.054	0.024	0.019	0.021	0.021	0.019
Not reverted	0.006	0.027	0.010	0.007	0.013	0.007	0.009
Bot	0.035			0.000		0.251	
One-click tool	0.028	0.121	0.093	0.099	0.096	0.096	0.083
Other registered, manual	0.035	0.094	0.059	0.020	0.025	0.046	0.087
Unregistered	0.024	0.134	0.073	0.046	0.035	0.058	0.085
Self	0.013	0.059	0.028	0.015	0.046	0.022	0.039

Table S6. Tag coverage: radius-rule reverts whose reverted first edit carries no mw-reverted tag, first edits from October 2020 (page-history samples)

Patrolling regime: English, Korean, Japanese none; German, Polish FlaggedRevs; Italian, French recent-changes patrolling. Cells: untagged share with 95 percent Wilson interval; number of reverts.

By class of reverter.

reverter_class	English	Italian*	Korean	German	Polish*	French	Japanese*
All reverters	0.003 [0.002, 0.004]; n = 16,611	0.096 [0.091, 0.101]; n = 14,935	0.004 [0.003, 0.005]; n = 9,485	0.024 [0.022, 0.027]; n = 12,246	0.039 [0.036, 0.043]; n = 13,282	0.070 [0.065, 0.075]; n = 10,914	0.010 [0.008, 0.013]; n = 7,039
Bot	0.000 [0.000, 0.003]; n = 2,012		0.000 [0.000, 0.071]; n = 50			0.004 [0.001, 0.024]; n = 231	
One-click tool	0.002 [0.001, 0.003]; n = 8,433	0.006 [0.005, 0.008]; n = 7,952	0.002 [0.001, 0.004]; n = 4,809	0.022 [0.018, 0.027]; n = 4,413	0.052 [0.045, 0.061]; n = 3,112	0.003 [0.002, 0.006]; n = 2,682	0.015 [0.010, 0.022]; n = 1,813
Other registered, manual	0.003 [0.002, 0.005]; n = 4,942	0.106 [0.099, 0.114]; n = 5,964	0.004 [0.002, 0.007]; n = 2,793	0.016 [0.013, 0.019]; n = 7,305	0.020 [0.017, 0.023]; n = 9,363	0.031 [0.027, 0.035]; n = 7,274	0.006 [0.004, 0.009]; n = 3,884
Unregistered	0.002 [0.000, 0.010]; n = 551	0.786 [0.745, 0.822]; n = 434	0.002 [0.000, 0.007]; n = 1,107	0.235 [0.179, 0.301]; n = 183	0.286 [0.245, 0.332]; n = 412	0.715 [0.653, 0.770]; n = 228	0.000 [0.000, 0.006]; n = 624
Self	0.022 [0.014, 0.036]; n = 673	0.708 [0.670, 0.744]; n = 579	0.022 [0.014, 0.035]; n = 726	0.127 [0.094, 0.170]; n = 299	0.135 [0.103, 0.174]; n = 364	0.733 [0.693, 0.770]; n = 499	0.027 [0.018, 0.042]; n = 697

Registered reverters (tool, other manual and self), by the user groups the reverting account held at the time of the revert (event_user_groups_historical). Autopatrol-carrying groups: autopatrolled, autoreviewer, sysop, bot, rollbacker, patroller, editor, reviewer and other administrative groups. Implicit groups such as autoconfirmed are not recorded in this field.

groups	English	Italian*	Korean	German	Polish*	French	Japanese*
autopatrol-carrying group	0.002 [0.001, 0.003]; n = 9,259	0.008 [0.006, 0.009]; n = 13,080	0.003 [0.002, 0.005]; n = 6,180	0.016 [0.014, 0.019]; n = 11,519	0.022 [0.020, 0.025]; n = 11,815	0.003 [0.002, 0.005]; n = 4,997	0.020 [0.015, 0.027]; n = 2,199
other group only	0.002 [0.001, 0.004]; n = 3,186		0.000 [0.000, 0.004]; n = 892	0.071 [0.035, 0.139]; n = 99	0.045 [0.025, 0.079]; n = 244	0.006 [0.002, 0.018]; n = 487	0.002 [0.001, 0.004]; n = 2,634
no group	0.012 [0.008, 0.018]; n = 1,603	0.698 [0.674, 0.722]; n = 1,408	0.013 [0.008, 0.021]; n = 1,256	0.143 [0.112, 0.181]; n = 399	0.163 [0.139, 0.190]; n = 780	0.117 [0.108, 0.126]; n = 4,971	0.013 [0.009, 0.020]; n = 1,561

Registered manual reverters other than the author with no explicit group, by the account's edit count at the time of the revert. On the French edition the autopatrolled status is granted automatically with experience and is not recorded as an explicit group.

reverter_edit_count	English	Italian*	Korean	German	Polish*	French	Japanese*
under 50	0.010 [0.003, 0.036]; n = 199	0.718 [0.610, 0.806]; n = 78	0.000 [0.000, 0.014]; n = 262		0.377 [0.272, 0.495]; n = 69	0.701 [0.608, 0.779]; n = 107	0.000 [0.000, 0.015]; n = 249
50 to 499	0.003 [0.001,	0.690 [0.606,	0.000 [0.000,		0.289 [0.205,	0.597 [0.524,	0.003 [0.001,

reverter_edit_count	English	Italian*	Korean	German	Polish*	French	Japanese*
	0.017]; n = 330	0.763]; n = 129	0.020]; n = 187		0.390]; n = 90	0.665]; n = 181	0.016]; n = 353
500 to 4,999		0.826 [0.780, 0.865]; n = 305			0.111 [0.060, 0.198]; n = 81	0.034 [0.024, 0.049]; n = 789	0.005 [0.001, 0.027]; n = 205
5,000 or more		0.755 [0.696, 0.806]; n = 233			0.133 [0.084, 0.206]; n = 120	0.001 [0.000, 0.003]; n = 3,274	

All reverters, by year of the first edit.

year	English	Italian*	Korean	German	Polish*	French	Japanese*
2020	0.003	0.090	0.000	0.016	0.021	0.071	0.008
2021	0.003	0.092	0.006	0.023	0.023	0.073	0.009
2022	0.003	0.088	0.005	0.026	0.039	0.078	0.016
2023	0.002	0.088	0.002	0.019	0.041	0.062	0.010
2024	0.004	0.093	0.004	0.038	0.054	0.070	0.005
2025	0.002	0.121	0.005	0.020	0.048	0.060	0.006
2026	0.002	0.141	0.003	0.015	0.067	0.089	0.015

Worked example: one cross-edition comparison computed with the tag and with the radius rule (first edits from October 2020; share of first edits counted as reverted; gap to English in percentage points; odds ratio of retention for reverted against non-reverted newcomers).

Edition	n	Reverted, tag	Reverted, radius rule	Gap to English, tag	Gap to English, radius rule	Change in gap [95% interval]	OR retention, tag	OR retention, radius rule
English	42,042	0.401	0.395	0	0		0.54	0.53
Italian*	41,609	0.327	0.359	7.4	3.6	3.8 [3.6, 4.0]	0.46	0.45
Korean	34,456	0.28	0.275	12.1	12	0.1 [0.0, 0.3]	0.48	0.48
German	42,644	0.282	0.287	11.9	10.8	1.1 [1.0, 1.3]	0.58	0.58
Polish*	42,613	0.303	0.312	9.9	8.3	1.5 [1.4, 1.7]	0.61	0.61
French	42,312	0.243	0.258	15.8	13.7	2.1 [1.9, 2.2]	0.61	0.6
Japanese*	42,442	0.171	0.166	23	22.9	0.1 [-0.1, 0.2]	0.46	0.46

Table S7. Registered tests in the held-out editions, as produced by the registered code

Bootstrap replicates in the registered code: 2,000 for H1 and H2, 500 for H4 (H4 intervals recomputed with 2,000 replicates are given below).

	Japanese*	Polish*	Italian*
n	60000.000	59997.000	59999.000
delta24_pp	2.972	6.217	15.900

	Japanese*	Polish*	Italian*
deltainf_pp	2.730	5.450	13.409
D_pp	0.242	0.767	2.492
D_ci90_lo	0.188	0.680	2.362
D_ci95_lo	0.180	0.660	2.338
D_ci95_hi	0.307	0.872	2.650
R	0.919	0.877	0.843
R_ci_lo	0.898	0.861	0.834
R_ci_hi	0.939	0.892	0.852
h2_n	60000.000	59997.000	59999.000
h2_delta_strict_pp	3.477	6.995	17.955
h2_delta_replace_pp	2.972	6.217	15.900
h2_S	0.145	0.111	0.114
h2_S_ci95_hi_onesided	0.157	0.119	0.119
h2_S_ci_lo	0.131	0.102	0.109
h2_S_ci_hi	0.160	0.120	0.121
h3_n	44211.000	44041.000	43006.000
h3_agree_d2_d3r15	0.986	0.979	0.954
h3_agree_d1_d2	0.981	0.977	0.951
h3_n_d1only	538.000	876.000	1997.000
h3_d1only_beyond_radius	0.424	0.094	0.073
h4_or_ratio	1.836	1.425	1.783
h4_or_ratio_ci_lo	1.791	1.375	1.720
h4_or_ratio_ci_hi	1.878	1.494	1.841
h4_boot	500.000	500.000	500.000

H4 recomputed with 2,000 paired bootstrap replicates, nine registered cells (ratio of the largest to the smallest odds ratio of retention across cells).

Edition	max/min odds ratio	95% interval	one-sided 95% lower bound
English	1.785	[1.768, 1.801]	1.771
Italian*	1.783	[1.722, 1.846]	1.732
Korean	3.008	[2.801, 3.221]	2.84
German	1.629	[1.592, 1.665]	1.599
Polish*	1.425	[1.368, 1.492]	1.379
French	1.512	[1.485, 1.538]	1.489
Japanese*	1.836	[1.791, 1.887]	1.798

H4 recomputed with 2,000 paired bootstrap replicates, three 24-hour cells (ratio of the largest to the smallest odds ratio of retention across cells).

Edition	max/min odds ratio	95% interval	one-sided 95% lower bound
English	1.492	[1.479, 1.504]	1.48
Italian*	1.503	[1.454, 1.552]	1.462
Korean	1.933	[1.835, 2.028]	1.851
German	1.488	[1.461, 1.516]	1.466
Polish*	1.302	[1.265, 1.344]	1.272
French	1.3	[1.283, 1.317]	1.285
Japanese*	1.321	[1.304, 1.339]	1.307

Table S8. Registered robustness analyses (bots and one-click tools excluded together, as registered)

Replace effect in percentage points by window (dump rule, page-history samples).

W	English	Italian*	Korean	German	Polish*	French	Japanese*
12 h	18.65	15.70	9.70	9.52	6.12	6.30	3.00
24 h	18.83	15.90	9.83	9.59	6.22	6.36	2.97
48 h	18.79	15.79	9.76	9.56	6.24	6.34	2.94
7 d	18.46	15.17	9.43	9.48	6.11	6.17	2.94
30 d	17.87	14.33	8.84	9.31	5.89	5.89	2.87
180 d	17.44	13.62	7.95	8.92	5.63	5.56	2.75
unbounded	17.37	13.41	7.62	8.79	5.45	5.45	2.73

Strict effect in percentage points by filter variant, 24 hours (full cohorts).

variant	English	Italian*	Korean	German	Polish*	French	Japanese*
F1 bots (current or historical)	4.53	0.01	0.07	0.04	0.03	1.02	0.01
F1c bots (current only)	4.53	0.01	0.03	0.04	0.02	1.02	0.01
F1fast bots + <=5 s registered	4.57	0.02	0.08	0.07	0.03	1.03	0.01
F1h bots (historical only)	4.53	0.01	0.07	0.04	0.03	1.02	0.01
F2 bots + tools (rollback/huggle/twinkle)	21.09	17.94	11.19	10.54	7.01	7.22	3.46
F2fast bots + tools + <=5 s	21.10	17.94	11.20	10.54	7.01	7.23	3.46
F2s bots + tools + undo	32.33	28.11	17.06	17.85	14.56	18.91	9.15
F3 bots + tools + anonymous reverters	21.92	18.77	13.27	10.96	7.68	7.53	4.57

Bot-or-tool share of first reverts within 24 hours by cohort variant.

cohort	English	Italian*	Korean	German	Polish*	French	Japanese*
all registered first edits 2019+	0.565	0.523	0.511	0.398	0.317	0.292	0.259
self-created first edits 2019+ (primary)	0.570	0.525	0.519	0.411	0.316	0.297	0.248
self-created, anchors 2021+ (all tags available)	0.649	0.546	0.555	0.395	0.305	0.282	0.301
self-created, secondary anchors (edits 2-5 within 91 d)	0.564	0.517	0.512	0.409	0.335	0.315	0.280

Korean, with and without the surge of June 2023 to December 2024.

Cohort	First edits	p reverted, 24 h	Bot-or-tool share, 24 h	p reverted, unbounded	R, dump rule
all	47633	0.216	0.519	0.279	0.775
excl. 2023-06..2024-12 surge	35065	0.165	0.437	0.226	0.865
surge only	12568	0.357	0.624	0.427	0.692

Leave-one-edition-out and by-year results are in 04_output/robust_dose_response_loo.csv and 04_output/<edition>_grid_by_year.csv.

Table S9. Change tags on reverting revisions by edition and year (all namespaces)

mw-rollback

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	1122426	895626	932095	883106	750250	774662	589221	606866	445436
Italian*	119804	124817	147566	159560	156810	127405	103429	90733	44813
Korean	20961	14881	28911	22528	31437	27802	29237	28939	11935
German	193873	189579	154538	142501	117173	120366	97322	69006	44425
Polish*	33863	33984	42665	37281	36801	33087	34604	22962	12179
French	82033	86959	68743	76641	67687	60521	59402	47767	33146
Japanese*	9121	11759	18030	27752	24273	26174	19051	14461	8951

huggle

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	345176	262033	162423	1631	2409	151939	132999	104257	47488
Italian*	0	0	0	0	0	0	0	0	0
Korean	0	0	0	0	0	0	0	0	0
German	0	0	0	0	0	0	0	0	0
Polish*	0	0	0	0	0	0	0	0	0
French	5016	4433	836	1111	1774	4027	1868	1510	280
Japanese*	0	0	0	0	0	0	0	0	0

twinkle

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	0	0	207362	602187	543831	514898	541969	538128	322779
Italian*	0	0	0	0	0	0	0	0	0
Korean	0	0	0	0	0	0	0	0	0
German	0	0	0	0	0	0	0	0	0
Polish*	0	0	0	0	0	0	0	0	0
French	0	0	0	0	0	0	0	0	0
Japanese*	0	0	0	0	0	0	0	0	0

mw-undo

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	1494822	1529089	1540633	1552495	1435631	1340784	1362885	1285981	803483
Italian*	95578	117098	116429	114948	97846	89979	84370	73064	45823
Korean	30946	29764	57931	45761	26982	21147	28007	21249	11311
German	109547	96185	98655	97514	81234	79920	75943	69172	44466
Polish*	24450	26055	33152	46991	49128	31284	32403	22045	13004
French	161866	154328	168055	151144	134478	137671	119977	118490	84144
Japanese*	74816	95772	95019	93695	94968	67232	50025	44876	31462

Table S10. Sensitivity of intervals to clustering in reverting accounts, and H3 under two windows (page-history samples)

Cells: estimate; 95 percent percentile interval resampling editors; 95 percent percentile interval resampling the accounts that made the first revert (non-reverted edits and edits first reverted by an unregistered editor are their own clusters; 1,000 replicates). No registered decision changes:

under account resampling the one-sided 95 percent lower bound of D is above 0 and the upper bound of S below .25 in the three held-out editions.

quantity	English	Italian*	Korean	German	Polish*	French	Japanese*
H1 difference D, dump rule (points)	1.463; editors [1.312, 1.610]; accounts [0.753, 2.405]	2.492; editors [2.343, 2.638]; accounts [1.875, 3.086]	2.213; editors [2.030, 2.400]; accounts [0.954, 3.587]	0.793; editors [0.695, 0.890]; accounts [0.476, 1.122]	0.767; editors [0.657, 0.873]; accounts [0.431, 1.115]	0.905; editors [0.815, 1.015]; accounts [0.543, 1.341]	0.242; editors [0.177, 0.305]; accounts [0.051, 0.469]
R, dump rule	0.922; editors [0.915, 0.930]; accounts [0.901, 0.949]	0.843; editors [0.835, 0.852]; accounts [0.828, 0.859]	0.775; editors [0.758, 0.791]; accounts [0.720, 0.838]	0.917; editors [0.907, 0.927]; accounts [0.901, 0.939]	0.877; editors [0.861, 0.894]; accounts [0.854, 0.911]	0.858; editors [0.842, 0.872]; accounts [0.837, 0.885]	0.919; editors [0.899, 0.939]; accounts [0.886, 0.974]
R, radius rule	1.045; editors [1.040, 1.050]; accounts [1.028, 1.067]	1.024; editors [1.020, 1.029]; accounts [1.016, 1.034]	1.074; editors [1.063, 1.085]; accounts [1.050, 1.111]	1.036; editors [1.031, 1.042]; accounts [1.022, 1.055]	1.067; editors [1.057, 1.077]; accounts [1.044, 1.099]	1.039; editors [1.031, 1.046]; accounts [1.023, 1.062]	1.048; editors [1.037, 1.060]; accounts [1.023, 1.098]
H2 share S (registered definition)	0.100; editors [0.095, 0.105]; accounts [0.083, 0.116]	0.114; editors [0.109, 0.120]; accounts [0.103, 0.124]	0.122; editors [0.113, 0.130]; accounts [0.104, 0.141]	0.089; editors [0.082, 0.096]; accounts [0.080, 0.098]	0.111; editors [0.102, 0.121]; accounts [0.101, 0.124]	0.120; editors [0.110, 0.129]; accounts [0.090, 0.154]	0.145; editors [0.129, 0.160]; accounts [0.128, 0.173]
Odds ratio of retention, tool-reverted vs not reverted	0.370; editors [0.343, 0.398]; accounts [0.342, 0.399]	0.306; editors [0.283, 0.331]; accounts [0.261, 0.353]	0.134; editors [0.115, 0.155]; accounts [0.095, 0.201]	0.283; editors [0.252, 0.314]; accounts [0.244, 0.331]	0.301; editors [0.265, 0.336]; accounts [0.265, 0.345]	0.284; editors [0.244, 0.328]; accounts [0.241, 0.327]	0.088; editors [0.067, 0.109]; accounts [0.061, 0.114]
Untagged share of radius-rule reverts	0.003; editors [0.002, 0.004]; accounts [0.002, 0.004]	0.096; editors [0.091, 0.101]; accounts [0.076, 0.123]	0.004; editors [0.003, 0.005]; accounts [0.003, 0.005]	0.024; editors [0.021, 0.027]; accounts [0.021, 0.028]	0.039; editors [0.036, 0.043]; accounts [0.031, 0.049]	0.070; editors [0.065, 0.074]; accounts [0.059, 0.084]	0.010; editors [0.008, 0.012]; accounts [0.007, 0.014]

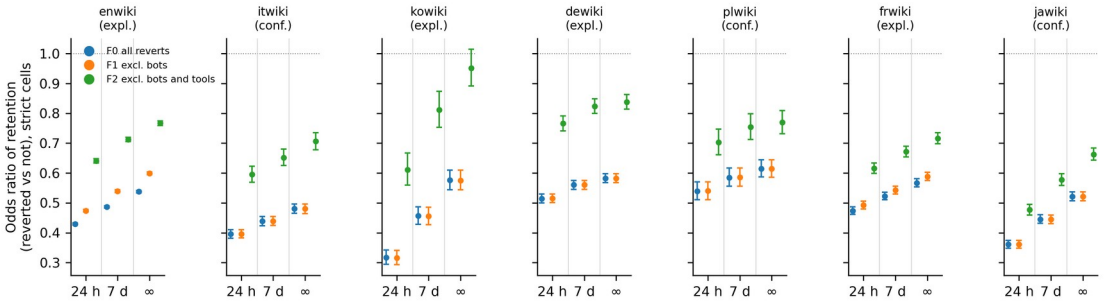
H3, second clause: of first edits reverted under the dump rule but not tagged, the share whose first revert lies beyond the 15-revision radius.

window	English	Italian*	Korean	German	Polish*	French	Japanese*
from 2020-10-01	0.841	0.088	0.954	0.137	0.131	0.114	0.749
registered window (from 2020-08-01)	0.340	0.073	0.770	0.075	0.094	0.096	0.424

Table S11. The proposed reporting items (Table 5 of the article) completed for this study

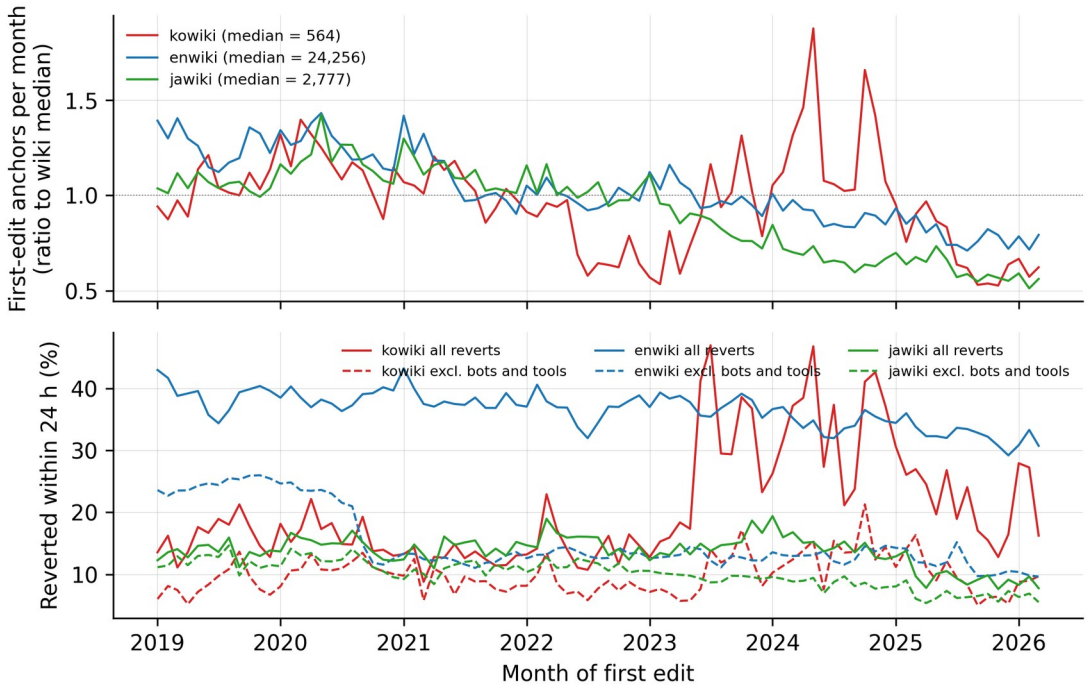
Item	This study
1. Construct	None adopted; the study compares what the indicator contains under each of four constructs (Table 1).
2. Attribution rule	Three, compared: the revision_is_identity_reverted field of the history dumps (unbounded); the same hashes with a 15-revision radius; the mw-reverted tag.
3. Window	24 hours, 7 days, 180 days and unbounded, all reported; 12 hours, 48 hours and 30 days in Table S8. No window is treated as validated for the non-English editions.
4. Bots, tools, author	Each counted and reported as a separate class; exclusions shown as variants (Table 3, Table S3).
5. Shares and accounts	Table 3 and Table S2, with the number of accounts making tool reverts and the share of the most active.
6. Outcomes by class	Retention by class of first reverter (Figure 3, Table S5).
7. Both rules	Follow-up and window results under the unbounded and the radius rule (Table 4, Table S4).
8. Tag	Patrolling regime and autopatrol rules per edition (Endnote 2); tag used for first edits from October 2020; coverage against the radius rule (Table S6).

Figure S1. Odds ratio of retention in the nine registered cells, by edition



Reverted versus non-reverted first edits under three reverter filters (all reverts; without bots; without bots and one-click tools, strict recoding) and three windows. The movement across cells is compositional (see the article).

Figure S2. Monthly newcomer first edits and 24-hour revert rates in Korean, English and Japanese, 2019 to March 2026



Upper panel: first edits per month relative to the edition's median. Lower panel: share reverted within 24 hours, all reverts (solid) and without bots and one-click tools (dashed). The drop of the dashed English line in September 2020 marks the start of Twinkle tagging.